Jiazheng Zhu

Ph.D. Candidate in Astrophysics

University of Science and Technology of China (USTC)

Hefei, Anhui, China

jiazheng@mail.ustc.edu.cn | +86-156-5657-3212

Homepage ORCID

Research Profile

I am a Ph.D. candidate in astrophysics at USTC, studying **time-domain astrophysics and optical transients**, with a particular focus on **tidal disruption events (TDEs)**. My research combines wide-field transient surveys, photometric and spectroscopic follow-up, and multiwavelength analysis to connect transient discovery with physical interpretation.

My future research aims to use TDEs to probe black holes across new mass and redshift regimes, particularly **intermediate-mass black holes (IMBHs)** and **high-redshift TDEs**. By combining deep surveys, including the Vera C. Rubin Observatory's Legacy Survey of Space and Time (LSST), with rapid follow-up, I aim to uncover faint and poorly sampled transient populations and investigate black-hole demographics and transient accretion physics.

Education

University of Science and Technology of China 2021–Present

Ph.D. Candidate in Astrophysics, School of Physical Sciences; Hefei, China

Integrated master's–doctoral program. Advisors: Prof. Tinggui Wang and Prof. Ning Jiang.

University of Science and Technology of China 2017–2021

B.S. in Astronomy, School of Physical Sciences; Hefei, China

Research Interests

- **Time-domain astrophysics and optical transients:** discovery and physical characterization through wide-field surveys and multiwavelength follow-up.
- **Tidal disruption events:** observational diversity, transient accretion, host-galaxy environments, and probes of otherwise quiescent black holes.
- **IMBHs and high-redshift TDEs:** extending black-hole searches to new mass, host-galaxy, and redshift regimes.
- **Faint and unexplored transient populations:** faint nuclear transients and unusual accretion events accessible to increasingly deep surveys.

Honors & Awards

National Scholarship for Graduate Students 2023

Ministry of Education of the People's Republic of China

Outstanding Volunteer 2023

USTC Summer Camp for Outstanding Undergraduates

Publications

Four first-author publications and 18 co-authored entries, including one correction. Paper titles link to their DOI records.

First-author Publications

[1] **Zhu, J.**, Xu, Z., Jiang, N., et al. (2026). **Discovery of a Featureless Tidal Disruption Event at $z \sim 1$ with the Wide Field Survey Telescope**. *The Astrophysical Journal Letters*, 1004, L10.

[2] **Zhu, J.**, Jiang, N., Wang, Y., et al. (2025). **Ultraviolet Spectral Evidence for Ansky as a Slowly Evolving Featureless Tidal Disruption Event with Quasiperiodic Eruptions**. *The Astrophysical Journal Letters*, 994, L16.

[3] **Zhu, J.**, Jiang, N., Wang, T., et al. (2023). **AT 2023clx: The Faintest and Closest Optical Tidal Disruption Event Discovered in Nearby Star-forming Galaxy NGC 3799**. *The Astrophysical Journal Letters*, 952, L35.

[4] **Zhu, J.**, Jiang, N., Dong, S., et al. (2023). **SN 2017egm: A Helium-rich Superluminous Supernova with Multiple Bumps in the Light Curves**. *The Astrophysical Journal*, 949, 23.

Co-author Publications

[5] Liu, H., Sun, L., Jiang, N., Shu, X., Wang, Y., Wang, T., Cutri, R. M., Dou, L., Zhang, F., **Zhu, J.**, et al. (2026). **An Obscured Tidal Disruption Event Uncovered by Its Mid- and Near-Infrared Dust Echo in a Star-forming Galaxy**. *The Astrophysical Journal*, 1005, 178.

[6] Li, D., Zhang, W., Yang, J., Chen, J.-H., Yuan, W., Cheng, H., Xu, F., Shu, X., Shen, R.-F., Jiang, N., **Zhu, J.**, et al. (2026). **A Fast Powerful X-ray Transient from Possible Tidal Disruption of a White Dwarf**. *Science Bulletin*, 71, 538–546.

[7] Wang, J., Huang, M., Xue, Y., Jiang, N., Huang, S., Wang, Y., **Zhu, J.**, et al. (2026). **A Tidal Disruption Event from an Intermediate-mass Black Hole Revealed by Comprehensive Multi-wavelength Observations**. *Nature Communications*, 17, 2007.

[8] Wu, M., Jiang, N., **Zhu, J.**, Luo, D., Dou, L., Wang, T. (2025). **A Torus Remnant Revealed by the Infrared Echo of Tidal Disruption Event AT 2019qiz: Implications for the Missing Energy and Quasiperiodic Eruption Formation**. *The Astrophysical Journal Letters*, 988, L77.

[9] Huang, S., Wang, T., Jiang, N., Shen, R.-F., Chen, Z., Wang, Y., **Zhu, J.**, et al. (2025). **Unveiling the Cosmic Dance of Repeated Nuclear Transient ASASSN-14ko: Insights from Multiwavelength Observations**. *The Astrophysical Journal*, 988, 237.

[10] Lin, J., Wang, T., Cai, M., et al. (2025). **Minute-cadence Observations of the Galactic Plane with the Wide Field Survey Telescope (WFST): Overview, Methodology, and Early Results**. *The Astrophysical Journal Supplement Series*, 278, 29.

[11] Jiang, N., Luo, D., **Zhu, J.**, Cutri, R. M. (2025). **The Extraordinary Long-lasting Infrared Echo of PS16dtm Reveals an Extremely Energetic Nuclear Outburst**. *The Astrophysical Journal Letters*, 980, L17.

[12] Liu, Y., Fan, L., Hu, L., Lu, J., Lu, Y., Xu, Z., **Zhu, J.**, et al. (2025). **Classification of Real and Bogus Transients Using Active Learning and Semi-supervised Learning**. *Astronomy & Astrophysics*, 693, A105.

Co-author Publications (continued)

[13] Jin, C.-C., Li, D.-Y., Jiang, N., Dai, L.-X., Cheng, H.-Q., **Zhu, J.**, et al. (2025). **An Intermediate-mass Black Hole Lurking in a Galactic Halo Caught Alive during Outburst**. *arXiv e-prints*, arXiv:2501.09580.

[14] Sun, L., Jiang, N., Dou, L., Shu, X., **Zhu, J.**, et al. (2024). **Recurring Tidal Disruption Events a Decade Apart in IRAS F01004-2237**. *Astronomy & Astrophysics*, 692, A262.

[15] Lin, Z., Jiang, N., Wang, T., Kong, X., Li, D., He, H., **Zhu, J.**, et al. (2024). **The Unluckiest Star: A Spectroscopically Confirmed Repeated Partial Tidal Disruption Event AT 2022dbl**. *The Astrophysical Journal Letters*, 971, L26.

[16] Wang, Y., Wang, T., Jiang, N., Zhang, X., **Zhu, J.**, et al. (2024). **ASASSN-18ap: A Dusty Tidal Disruption Event Candidate with an Early Bump in the Light Curve**. *The Astrophysical Journal*, 966, 136.

[17] Huang, S., Jiang, N., Lin, Z., **Zhu, J.**, Wang, T. (2024). **Correction to: AT2018dyk Revisited: A Tidal Disruption Event Candidate with Prominent Infrared Echo and Delayed X-ray Emission in a LINER Galaxy**. *Monthly Notices of the Royal Astronomical Society*, 531, 4793.

[18] Huang, S., Jiang, N., **Zhu, J.**, Wang, Y., Wang, T., et al. (2024). **AT 2023lli: A Tidal Disruption Event with Prominent Optical Early Bump and Delayed Episodic X-Ray Emission**. *The Astrophysical Journal Letters*, 964, L22.

[19] Jiang, N., Zhou, Z., **Zhu, J.**, Wang, Y., Wang, T. (2023). **Two Candidate Obscured Tidal Disruption Events Coincident with High-energy Neutrinos**. *The Astrophysical Journal Letters*, 953, L12.

[20] Lin, Z., Jiang, N., Kong, X., Huang, S., Lin, Z., **Zhu, J.**, et al. (2022). **The Luminosity Function of Tidal Disruption Flares for the ZTF-I Survey**. *The Astrophysical Journal Letters*, 939, L33.

[21] Dou, L., Jiang, N., Wang, T., Shu, X., Yang, H., Pan, Z., **Zhu, J.**, et al. (2022). **X-ray View of a Merging Supermassive Black Hole Binary Candidate SDSS J1430+2303: Results from the First ∼200 Days of Observations**. *Astronomy & Astrophysics*, 665, L3.

[22] Wang, Y., Jiang, N., Wang, T., **Zhu, J.**, et al. (2022). **Discovery of ATLAS17jrp as an Optical-, X-Ray-, and Infrared-bright Tidal Disruption Event in a Star-forming Galaxy**. *The Astrophysical Journal Letters*, 930, L4.

Conference Presentations

Tidal Disruption Events and Nuclear Transients: Entering the Data-Rich Era
Heraklion, Crete, Greece 2–6 September 2024

- **Oral presentation:** "Mid-Infrared Outbursts in Nearby Galaxies (MIRONG)."
- **Poster:** AT 2023clx as a representative of the faint TDE population.

Observational Experience & Telescope Time

Observing Experience

- **Wide Field Survey Telescope (WFST), 2023–Present.** One week of on-site observing and six weeks of remote observing in total. Experience includes observing assistance, monitoring observing conditions, survey operations, and real-time image-quality assessment.
- **Lijiang 2.4-m Telescope, 2024.** Two nights of on-site observing.

Principal Investigator Programs

- **Gemini North, 2024B** `GN-2024B-DD-105`
  "Towards the first optical discovery of an IMBH-TDE at its Eddington accretion phase in a dwarf starburst galaxy."
  Approved time: **2 hours**.
- **Swift UVOT/XRT, 2023–2026.** Target-of-opportunity requests for individual transients. Total approved time: **129 ks**.

Co-Investigator Programs

- **JWST** `JWST-DD-12569`
  "Are LFBOTs Dusty? Constraining the Nature of IR Emission in the Luminous Fast Blue Optical Transient AT2026dbl."
  Approved time: **5.2 hours**.
- **Gemini North, 2024A** `GN-2024A-DD-103`
  "Towards a Deep Spectroscopic View of the First X-Ray IMBH-TDE Candidate Discovered at Early Phase."
  PI: Ning Jiang. Approved time: **3 hours**.
- **Gemini North, 2023A** `GN-2023A-Q-110`
  "Spectroscopic Monitoring of a SMBH Binary Candidate at its Last Inspiraling Stage."
  PI: Ning Jiang. Approved time: **4.5 hours**.
- **Gemini North, 2022A** `GN-2022A-DD-101`
  "Spectroscopic Diagnosis of Optical flares from a merging supermassive black hole binary."
  PI: Zhen Pan. Approved time: **2 hours**.
- **HST** `HST-GO/DD-18213`
  "Imaging of EP241021a during the flattening in the optical light curve: an exotic gamma-ray burst or a peculiar jetted tidal disruption event?"
  Approved time: **2 orbits**.
- **HST** `HST-GO/DD-17933`
  "Turn-on AGN or IMBH-TDE? UV Spectroscopic Observations May Tell."
  Approved time: **4 orbits**.
- **HST** `HST-GO/DD-17849`
  "Probing the UV spectral evolution of a unique TDE with early bright X-ray emission."
  Approved time: **4 orbits**.
- **HST** `HST-GO/DD-17609`
  "A UV spectroscopic view of the 2024 outburst of AT2020afhd: A Unique Opportunity to Test the Reprocessing Model of Tidal Disruption Event."
  Approved time: **2 orbits**.
- **VLA, 2026A** `VLA/26A-638`
  "VLA Observation of a Unique Off-nuclear TDE Candidate."
  Director's Discretionary Time; proposal date: 2 June 2026. Approved time: **2 hours at Priority B**, plus **8 hours at Priority C** conditional on detection in the initial observation.

- **LCOGT, 2023A** `NAOC2023A-005`
  "Continuous Daily Optical Monitoring of a SMBH Binary Candidate at its Last Inspiraling Stage."
  PI: Ning Jiang. Approved time: **35 hours**.
- **LCOGT, 2022B** `NAOC2022B-002`
  "Further Daily Optical Monitoring of an unprecedented SMBH Binary at its Last Inspiraling Stage."
  PI: Ning Jiang. Approved time: **20 hours**.
- **MMT/Binospec, 2025B** `UAO-S111-25B`
  "Spectroscopic Identification of Candidate Tidal Disruption Events Caused by Intermediate-Mass Black Holes."
  PI: Ann Zabludoff. Approved time: **1.5 nights** of queue observing.

Additional collaborative observing proposals include P200, Swift, XMM-Newton, and the Lijiang 2.4-m telescope; program details are not listed here.

Research & Technical Skills

- **Surveys and transient discovery:** transient detection and classification, survey data mining, candidate selection, light-curve analysis, variability characterization, and multiband photometry.
- **Imaging and photometry:** CCD calibration, image stacking, astrometry, image subtraction, difference imaging, and photometric measurements.
- **Spectroscopy:** optical/UV spectral analysis, continuum and emission-line characterization, and transient classification.
- **SED and physical modeling:** multiwavelength spectral energy distribution fitting, blackbody and continuum modeling, Markov chain Monte Carlo (MCMC), and Bayesian inference.
- **Host galaxies:** galaxy photometry and morphology, stellar-population analysis, host-galaxy SED fitting, and spectral fitting.
- **Scientific computing:** astronomical data analysis, scientific computing, and visualization, primarily in Python.

Teaching Experience

Teaching Assistant, Mechanics A 2023
School of Physical Sciences, University of Science and Technology of China

Teaching Assistant, Thermal Physics A 2021
School of Physical Sciences, University of Science and Technology of China

Service & Outreach

Astronomy Summer Camp Volunteer 2023, 2026
University of Science and Technology of China

Astronomy Association Member
Wenzhou, China (my hometown)

Campus Service & Astronomy Outreach
University of Science and Technology of China

Actively participate in university and astronomy department student activities and promote astronomy through science communication and outreach.